

DUSHYANT JOSHI

AI/ML Engineer | Generative AI | RAG | Agentic Systems | LLM Evaluation | LLMOps
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PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years shipping production Generative AI systems — Retrieval-Augmented Generation (RAG), multi-agent and Agentic AI workflows, LLM evaluation, and async Python services on FastAPI. Built an enterprise AI Copilot serving **1,500+ weekly production queries** at **sub-200ms retrieval** over **10,000+ documents**, plus the Langfuse observability and cost-analytics layer behind it. Owns the full lifecycle: retrieval design, agent orchestration, guardrails, evaluation and Docker/AWS deployment.

KEY HIGHLIGHTS

- **70% lift in customer engagement** from production RAG on LangChain, GPT-4o and Pinecone at sub-200ms retrieval.
- **40% cut in debugging time** via end-to-end Langfuse tracing across a 5-stage multi-agent chain.
- **60% faster query resolution** from a LangGraph agent reasoning over SQL and unstructured data.
- Cost analytics exposed **20% of queries driving 60% of compute**, triggering a model-routing strategy.

KEY SKILLS

Programming: Python, SQL, OOP, Data Structures & Algorithms, Git, Linux

Generative AI & Agents: RAG, Agentic AI, Multi-Agent Systems, ReAct Agents, Function/Tool Calling, Prompt Engineering, Few-Shot Learning, Fine-Tuning, Guardrails, Knowledge Graphs, MCP

LLM Frameworks & Models: LangChain, LangGraph, OpenAI API, Anthropic Claude API, GPT-4o, Llama-3.2, LiteLLM

LLM Evaluation: Offline & Online Evals, LLM-as-a-Judge, G-Eval, RAG Triad, Faithfulness, Answer Relevance, Recall@k, NDCG, Hallucination Testing, Regression & Safety Evals

Retrieval & Databases: Pinecone, Qdrant, FAISS, Vector Embeddings, Semantic & Hybrid Search, BM25, Reciprocal Rank Fusion, Reranking, PostgreSQL, MySQL, MongoDB

Backend & LLMops: FastAPI, Django, REST APIs, Microservices, Async Python, Docker, CI/CD, AWS, Langfuse, OpenTelemetry, Prompt Versioning, Cost Analytics

PROFESSIONAL EXPERIENCE

Analyttica Datalab | AI/ML Engineer

Mar 2025 – Present | Bengaluru, India

- **Engineered** core components of **Copilot**, an enterprise AI assistant unifying processing, inference and recommendation engines, and shipped **Agentic AI workflows** on **ReAct** architecture with planners, tools and memory serving **1,500+ weekly production queries**.
- **Architected** production **RAG pipelines** and an API integration router with **LangChain, GPT-4o and Pinecone**, driving a **70% increase in customer engagement** at **sub-200ms retrieval latency** across **10,000+ documents**.
- **Owned Langfuse** observability across a **5-stage multi-agent chain** (orchestrator, grounding, SQL generation, validation, response), instrumenting prompts, tokens, latency and cost per trace to **reduce debugging time 40%**.
- **Built** an AI usage and cost-analytics platform end to end — Langfuse ingestion to **MySQL** fact tables to layered **FastAPI** APIs; analysis revealed **20% of queries consumed 60% of compute**, driving model routing.
- **Designed** an enterprise-scale **KPI Knowledge Graph** unifying SQL, API and unstructured sources to automate recurring analytics workflows.
- **Developed** async REST microservices in **FastAPI** with **MongoDB** session management for multi-turn context, containerized via **Docker**; **delivered 5+ enterprise POCs** using few-shot learning, function calling and fine-tuning.

American Chase | Software Developer

Nov 2024 – Mar 2025 | Indore, India

- **Delivered** Python backend services for an in-house payroll and timesheet platform covering employee, attendance and payroll-processing workflows in an **Agile** team, spanning debugging, testing, Git-based development and deployment.

PROJECTS

Pro-RAG — Production RAG System

Python, FastAPI, Qdrant, PostgreSQL, LiteLLM

- **Engineered** a RAG pipeline with **hybrid retrieval** (dense vectors + BM25 fused by **reciprocal rank fusion**), **BGE reranking** and **HHEM-2.1** entailment verification, architected for **10M documents at 50 QPS**.
- **Defined** the eval strategy — **Recall@k/NDCG** for retrieval, faithfulness and answer relevance for generation — targeting **0.90 faithfulness** and **under 5% hallucination** with regression gating; **implemented** prompt-injection detection, **PII guardrails** and confidence-based refusal.

Multi-Modal Intelligent Query Agent

Python, LangGraph, RAG, FastAPI, PostgreSQL

- **Built** an autonomous agentic system with **LangGraph** using tool calling, planning, memory and multi-step reasoning over SQL data and unstructured documents; the async FastAPI backend **cut query-resolution time 60%**.

CERTIFICATIONS & ACHIEVEMENTS

- **Anthropic Academy (Claude)** — **5 course badges** (Sept 2026): Claude 101, Claude Platform 101, Claude Code 101, Claude Code in Action, Introduction to Claude Cowork; preparing for Claude certification. **HackerRank SQL Certified**.
- Self-directed study in **LLM/RAG evaluation**: offline and online evals, LLM-as-a-Judge, G-Eval, RAG Triad, retriever and generator testing, safety and regression evals.
- Solved **350+** LeetCode/GeeksforGeeks problems (algorithms, SQL). **National-level Volleyball player**.

EDUCATION

B.E. Electronics and Instrumentation, Bangalore Institute of Technology | CGPA 7.0/10

2020 – 2024

Python and AI/ML Training, PySpiders | Python, OOP, Django and FastAPI REST APIs

Feb 2024 – Oct 2024